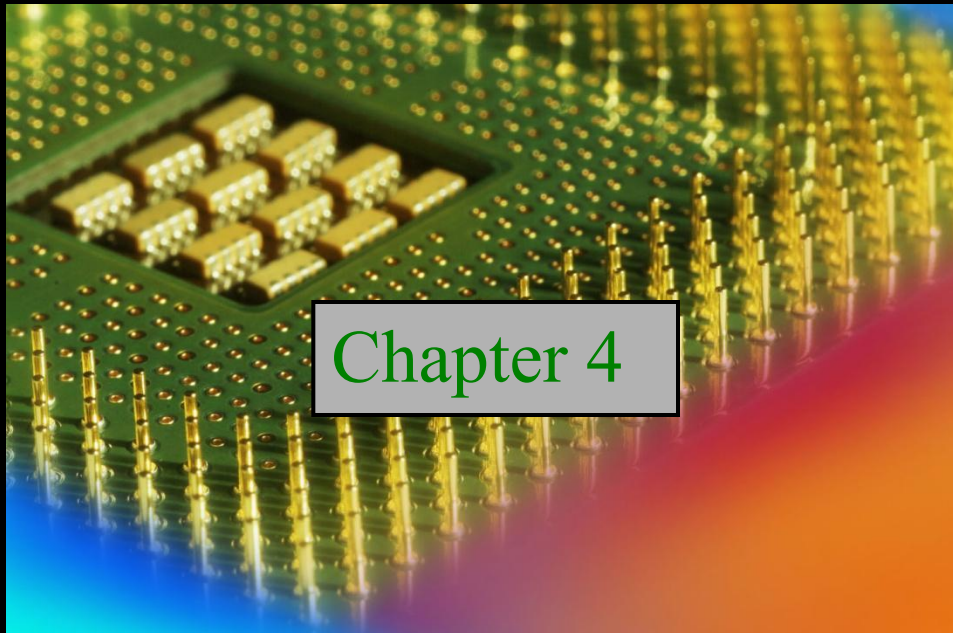


Digital Fundamentals

Tenth Edition

Floyd



Chapter 4

Boolean algebra and logic simplification

1. Boolean operations and expressions
2. Laws and rules of boolean algebra
3. De Morgan's theorems
4. Boolean analysis of logic circuits
5. Simplification using boolean algebra
6. Standard forms of boolean expressions
7. Boolean expressions and truth tables
8. The karnaugh map
9. Five-variable karnaugh maps



Boolean Addition

In Boolean algebra, a **variable** is a symbol used to represent an action, a condition, or data. A single variable can only have a value of 1 or 0.

The **complement** represents the inverse of a variable and is indicated with an overbar. Thus, the complement of A is $\overline{A}$.

A **literal** is a variable or its complement.

Addition is equivalent to the OR operation. The sum term is 1 if one or more of the literals are 1. The sum term is zero only if each literal is 0.

Example Determine the values of A , B , and C that make the sum term of the expression $\overline{A} + B + \overline{C} = 0$?

Solution Each literal must = 0; therefore $A = 1$, $B = 0$ and $C = 1$.

Boolean operations and expressions

Summary

Boolean Multiplication

In Boolean algebra, multiplication is equivalent to the AND operation. The product of literals forms a product term. The product term will be 1 only if all of the literals are 1.

Example What are the values of the A , B and C if the product term of $A \cdot \overline{B} \cdot \overline{C} = 1$?

Solution Each literal must = 1; therefore $A = 1$, $B = 0$ and $C = 0$.

Commutative Laws

The **commutative laws** are applied to addition and multiplication. For addition, the commutative law states
In terms of the result, the order in which variables are ORed makes no difference.

$$A + B = B + A$$

For multiplication, the commutative law states
In terms of the result, the order in which variables are ANDed makes no difference.

$$AB = BA$$

Summary

Associative Laws

The **associative laws** are also applied to addition and multiplication. For addition, the associative law states

When ORing more than two variables, the result is the same regardless of the grouping of the variables.

$$A + (B + C) = (A + B) + C$$

For multiplication, the associative law states

When ANDing more than two variables, the result is the same regardless of the grouping of the variables.

$$A(BC) = (AB)C$$

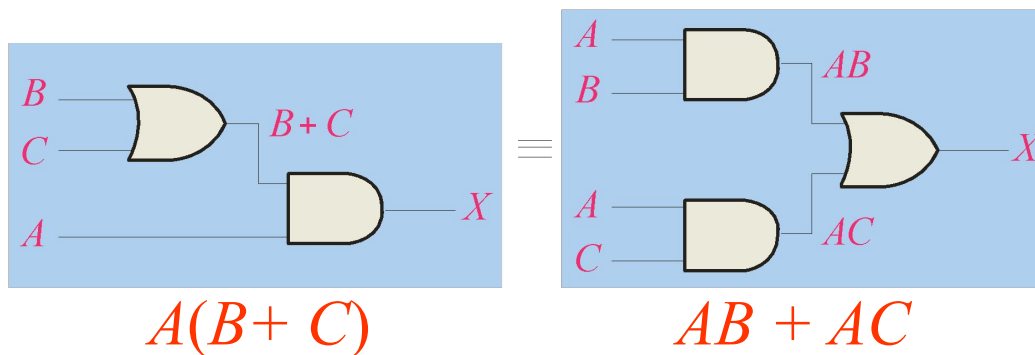
Summary

Distributive Law

The **distributive law** is the factoring law. A common variable can be factored from an expression just as in ordinary algebra. That is

$$AB + AC = A(B + C)$$

The distributive law can be illustrated with equivalent circuits:



Rules of Boolean Algebra

$$1. A + 0 = A$$

$$2. A + 1 = 1$$

$$3. A \cdot 0 = 0$$

$$4. A \cdot 1 = A$$

$$5. A + A = A$$

$$6. A + \bar{A} = 1$$

$$7. A \cdot A = A$$

$$8. A \cdot \bar{A} = 0$$

$$9. \bar{\bar{A}} = A$$

$$10. A + AB = A$$

$$11. A + \bar{A}B = A + B$$

$$12. (A + B)(A + C) = A + BC$$

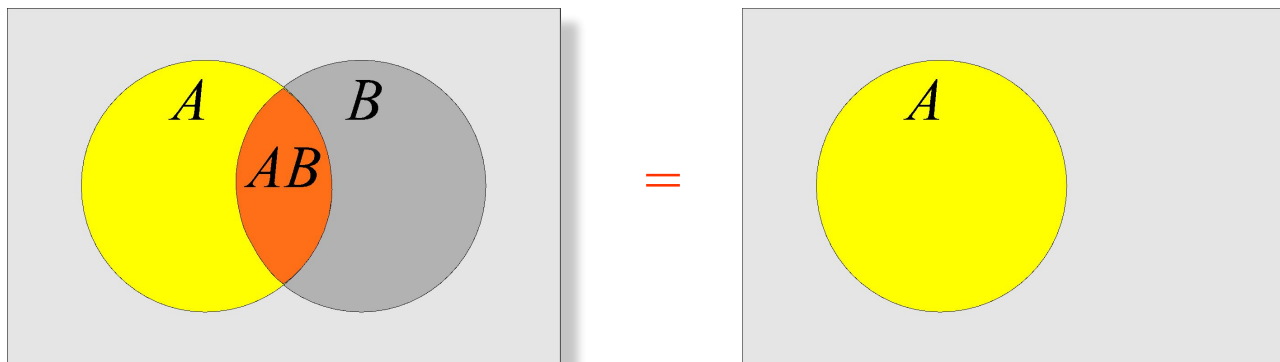
Summary

Rules of Boolean Algebra

Rules of Boolean algebra can be illustrated with *Venn* diagrams. The variable A is shown as an area.

The rule $A + AB = A$ can be illustrated easily with a diagram. Add an overlapping area to represent the variable B .

The overlap region between A and B represents AB .



The diagram visually shows that $A + AB = A$. Other rules can be illustrated with the diagrams as well.

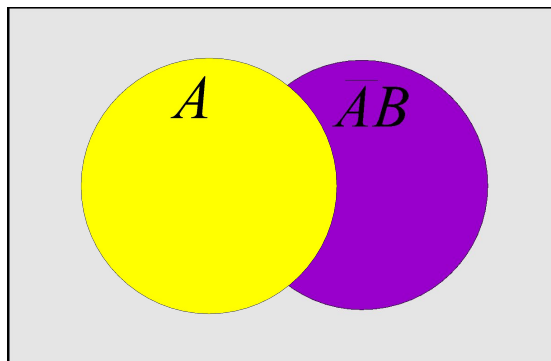
Summary

Rules of Boolean Algebra

Example Illustrate the rule $A + \bar{A}B = A + B$ with a Venn diagram.

Solution

This time, $\bar{A}$ is represented by the blue area and B again by the red circle. The intersection represents $\bar{A}B$. Notice that $A + \bar{A}B = A + B$



Summary

Rules of Boolean Algebra

Rule 12, which states that $(A + B)(A + C) = A + BC$, can be proven by applying earlier rules as follows:

$$\begin{aligned}(A + B)(A + C) &= AA + AC + AB + BC \\&= A + AC + AB + BC \\&= A(1 + C + B) + BC \\&= A \cdot 1 + BC \\&= A + BC\end{aligned}$$

This rule is a little more complicated, but it can also be shown with a Venn diagram, as given on the following slide...

Summary

Three areas represent the variables A , B , and C .

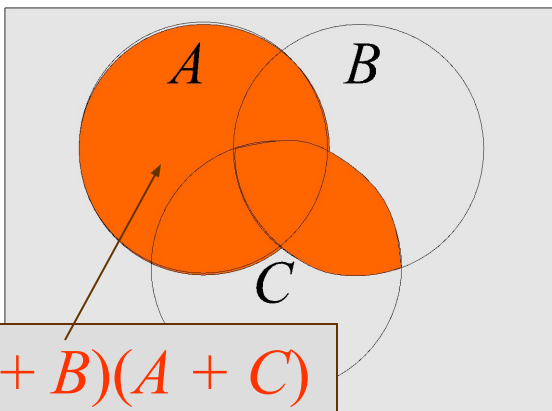
The area representing $A + B$ is shown in yellow.

The area representing $A + C$ is shown in red.

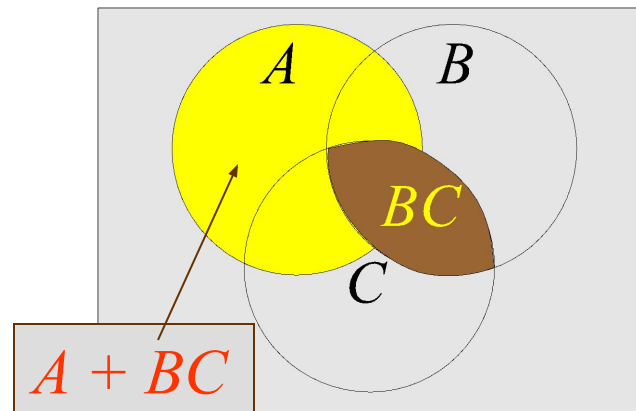
The overlap of red and yellow is shown in orange.

The overlapping area between B and C represents BC .

ORing with A gives the same area as before.



=



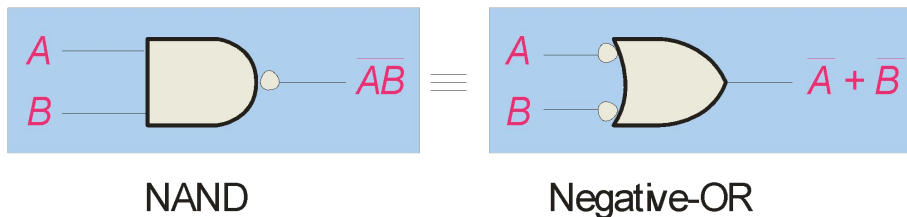
DeMorgan's Theorem

DeMorgan's 1st Theorem

The complement of a product of variables is equal to the sum of the complemented variables.

$$\overline{AB} = \overline{A} + \overline{B}$$

Applying DeMorgan's first theorem to gates:



Inputs		Output	
A	B	$\overline{AB}$	$\overline{A} + \overline{B}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

Summary

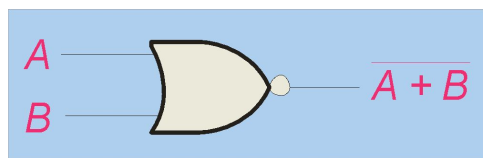
DeMorgan's Theorem

DeMorgan's 2nd Theorem

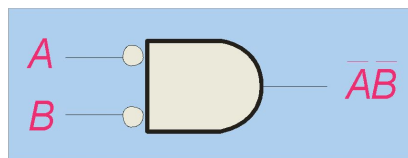
The complement of a sum of variables is equal to the product of the complemented variables.

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

Applying DeMorgan's second theorem to gates:



NOR



Negative-AND

Inputs		Output	
A	B	$\overline{A + B}$	$\overline{AB}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

Summary

DeMorgan's Theorem

Example Apply DeMorgan's theorem to remove the overbar covering both terms from the expression $X = \overline{C + D}$.

Solution To apply DeMorgan's theorem to the expression, you can break the overbar covering both terms and change the sign between the terms. This results in $X = \overline{\overline{C}} \cdot \overline{\overline{D}}$. Deleting the double bar gives $X = C \cdot \overline{D}$.

Boolean Analysis of Logic Circuits

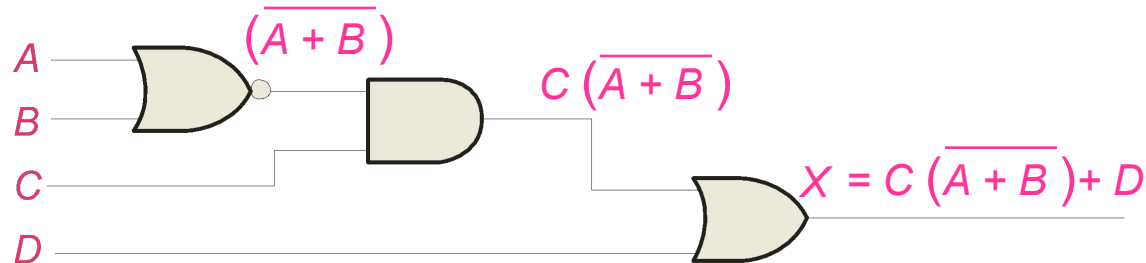
Combinational logic circuits can be analyzed by writing the expression for each gate and combining the expressions according to the rules for Boolean algebra.

Example

Apply Boolean algebra to derive the expression for X .

Solution

Write the expression for each gate:



Applying DeMorgan's theorem and the distribution law:

$$X = C (\overline{A} \overline{B}) + D = \overline{A} \overline{B} C + D$$

Summary

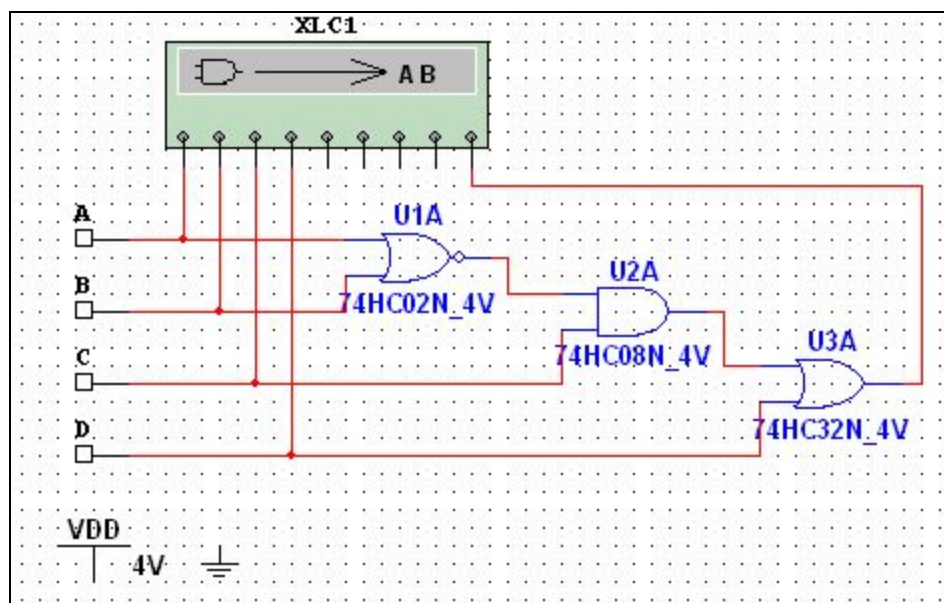
Boolean Analysis of Logic Circuits

Example

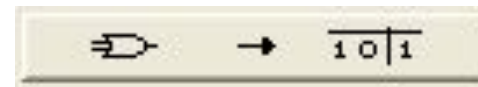
Use Multisim to generate the truth table for the circuit in the previous example.

Solution

Set up the circuit using the Logic Converter as shown. (Note that the logic converter has no “real-world” counterpart.)



Double-click the Logic Converter top open it. Then click on the conversion bar on the right side to see the truth table for the circuit (see next slide).



Summary

Boolean Analysis of Logic Circuits

The simplified logic expression can be viewed by clicking

$\overline{101} \xrightarrow{\text{SIMP}} A \overline{B}$

Simplified
expression

Logic Converter-XLC1

Out ☐

☐ A ☐ B ☐ C ☐ D ☐ E ☐ F ☐ G ☐ H

000	0	0	0	0					0
001	0	0	0	1					1
002	0	0	1	0					1
003	0	0	1	1					1
004	0	1	0	0					0
005	0	1	0	1					1
006	0	1	1	0					0
007	0	1	1	1					1
008	1	0	0	0					0
009	1	0	0	1					1
010	1	0	1	0					0
011	1	0	1	1					1
012	1	1	0	0					0
013	1	1	0	1					1
014	1	1	1	0					0
015	1	1	1	1					1

Conversions

$\Rightarrow \rightarrow \overline{101}$

$\overline{101} \rightarrow A \overline{B}$

$\overline{101} \xrightarrow{\text{SIMP}} A \overline{B}$

$A \overline{B} \rightarrow \overline{101}$

$A \overline{B} \rightarrow \Rightarrow$

$A \overline{B} \rightarrow \text{NAND}$

A'B'C+D

4-5 Simplification Using Boolean Algebra

- 1. $AB + A(B+C) + B(B+C)$
- $(AB + AC)' + A'B'C$

SOP and POS forms

Boolean expressions can be written in the **sum-of-products** form (**SOP**) or in the **product-of-sums** form (**POS**). These forms can simplify the implementation of combinational logic, particularly with PLDs. In both forms, an overbar cannot extend over more than one variable.

An expression is in SOP form when two or more product terms are summed as in the following examples:

$$\bar{A} \bar{B} \bar{C} + A B \quad A B C + \bar{C} \bar{D} \quad \bar{C} \bar{D} + E$$

An expression is in POS form when two or more sum terms are multiplied as in the following examples:

$$(A + B)(\bar{A} + C) \quad (A + B + C)(\bar{B} + D) \quad (A + B)C$$

Summary

SOP Standard form

In **SOP standard form**, every variable in the domain must appear in each term. This form is useful for constructing truth tables or for implementing logic in PLDs.

You can **expand** a nonstandard term to standard form by **multiplying** the term by a **term** consisting of the **sum of the missing variable and its complement**.

Example Convert $X = \bar{A} \bar{B} + A B C$ to standard form.

Solution The first term does not include the variable C . Therefore, multiply it by the $(C + \bar{C})$, which = 1:

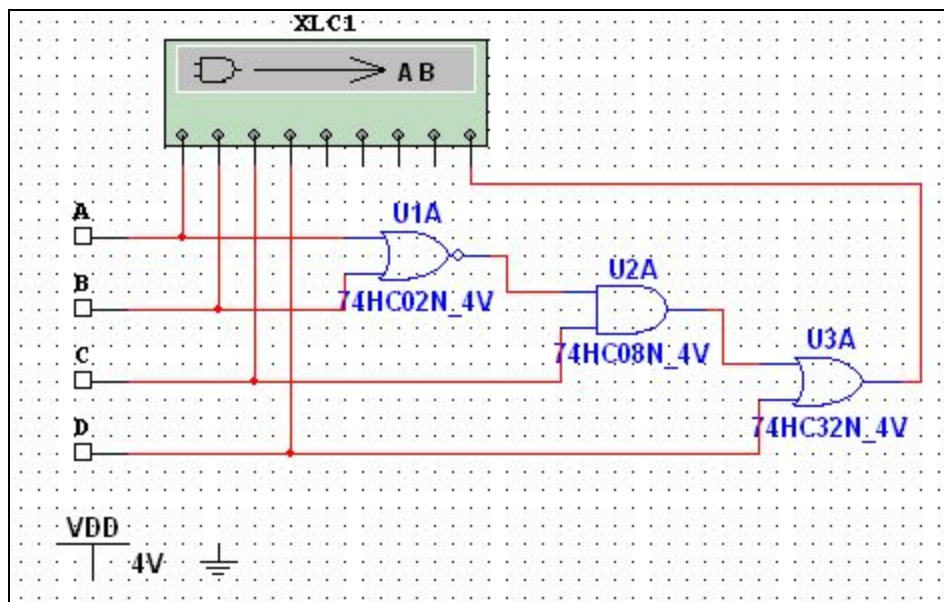
$$\begin{aligned} X &= \bar{A} \bar{B} (C + \bar{C}) + A B C \\ &= \bar{A} \bar{B} C + \bar{A} \bar{B} \bar{C} + A B C \end{aligned}$$

Summary

SOP Standard form

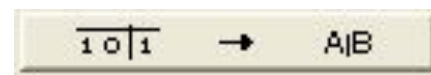
The Logic Converter in Multisim can convert a circuit into standard SOP form.

Example Use Multisim to view the logic for the circuit in standard SOP form.



Solution

Click the truth table to logic button on the Logic Converter.



See next slide...

Summary

SOP Standard form

SOP
Standard
form →

Logic Converter-XLC1

Out

A B C D E F G H

000	0	0	0	0				0
001	0	0	0	1				1
002	0	0	1	0				1
003	0	0	1	1				1
004	0	1	0	0				0
005	0	1	0	1				1
006	0	1	1	0				0
007	0	1	1	1				1
008	1	0	0	0				0
009	1	0	0	1				1
010	1	0	1	0				0
011	1	0	1	1				1
012	1	1	0	0				0
013	1	1	0	1				1
014	1	1	1	0				0
015	1	1	1	1				1

Conversions

$\Rightarrow$ → $\overline{1}0\overline{1}$

$\overline{1}0\overline{1}$ → A|B

$\overline{1}0\overline{1}$ $\xrightarrow{\text{SIMP}}$ A|B

A|B → $\overline{1}0\overline{1}$

A|B → $\Rightarrow$

A|B → NAND

Summary

POS Standard form

In **POS standard form**, every variable in the domain must appear in each sum term of the expression.

You can **expand** a nonstandard POS expression to standard form by **adding the product of the missing variable and its complement** and applying rule 12, which states that $(A + B)(A + C) = A + BC$.

Example Convert $X = (\bar{A} + \bar{B})(A + B + C)$ to standard form.

Solution The first sum term does not include the variable C . Therefore, add $C \bar{C}$ and expand the result by rule 12.

$$\begin{aligned} X &= (\bar{A} + \bar{B} + C \bar{C})(A + B + C) \\ &= (\bar{A} + \bar{B} + C)(\bar{A} + \bar{B} + \bar{C})(A + B + C) \end{aligned}$$

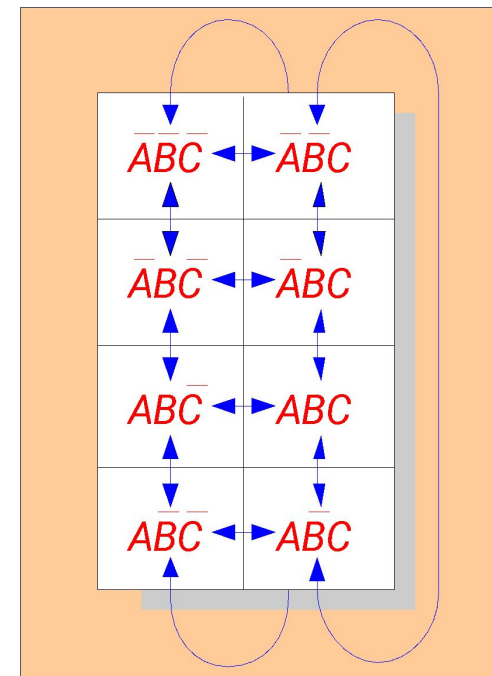
4-7 boolean expressions and truth tables

- Truth table = a common way of presenting a concise format of the logical operation of the circuit.
- Example.
- 1. develop a truth table for the standard SOP
 $A'B'C + AB'C' + ABC$
- 2. determine the truth table for the standard POS
 $(A+B+C)(A+B'+C)(A+B'+C')(A'+B+C')(A'+B'+C)$

Karnaugh maps

The Karnaugh map (K-map) is a tool for simplifying combinational logic with 3 or 4 variables. For 3 variables, 8 cells are required (2^3).

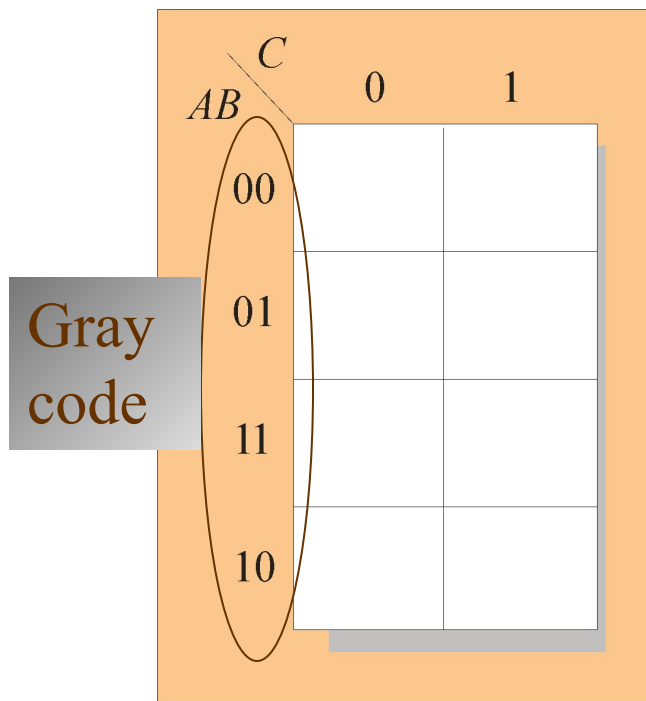
The map shown is for three variables labeled A , B , and C . Each cell represents one possible product term. Each cell differs from an adjacent cell by only one variable.



Summary

Karnaugh maps

Cells are usually labeled using 0's and 1's to represent the variable and its complement.



The numbers are entered in gray code, to force adjacent cells to be different by only one variable.

Ones are read as the true variable and zeros are read as the complemented variable.

Summary

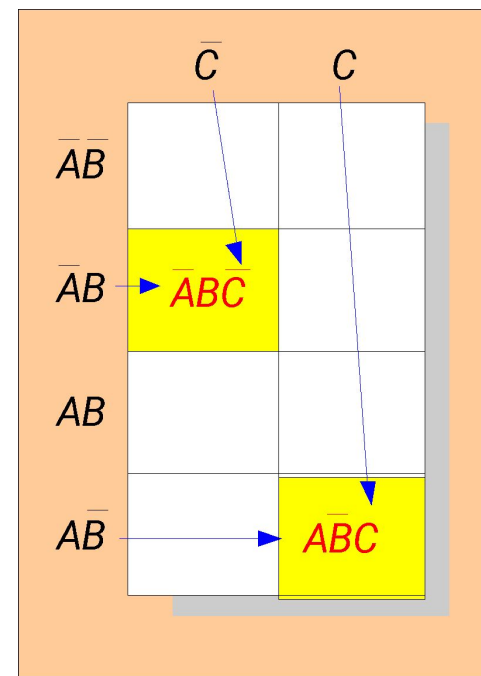
Karnaugh maps

Alternatively, cells can be labeled with the variable letters. This makes it simple to read, but it takes more time preparing the map.

Example Read the terms for the yellow cells.

Solution

The cells are $\bar{A}\bar{B}\bar{C}$ and $\bar{A}BC$.



Summary

Karnaugh maps

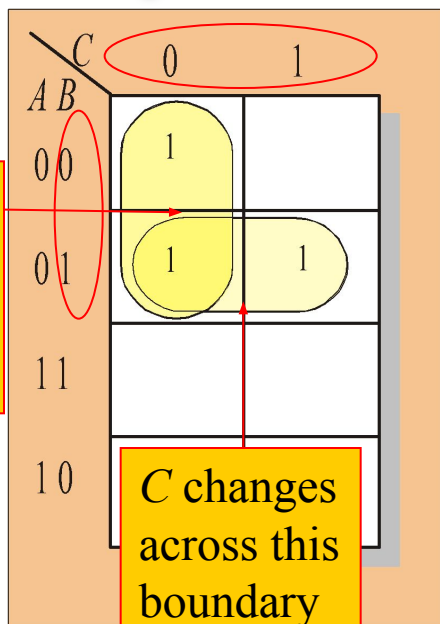
K-maps can simplify combinational logic by grouping cells and eliminating variables that change.

Example Group the 1's on the map and read the minimum logic.

Solution

1. Group the 1's into two overlapping groups as indicated.
2. Read each group by eliminating any variable that changes across a boundary.
3. The vertical group is read $\overline{A}C$.
4. The horizontal group is read $\overline{A}B$.

$$X = \overline{A}C + \overline{A}B$$



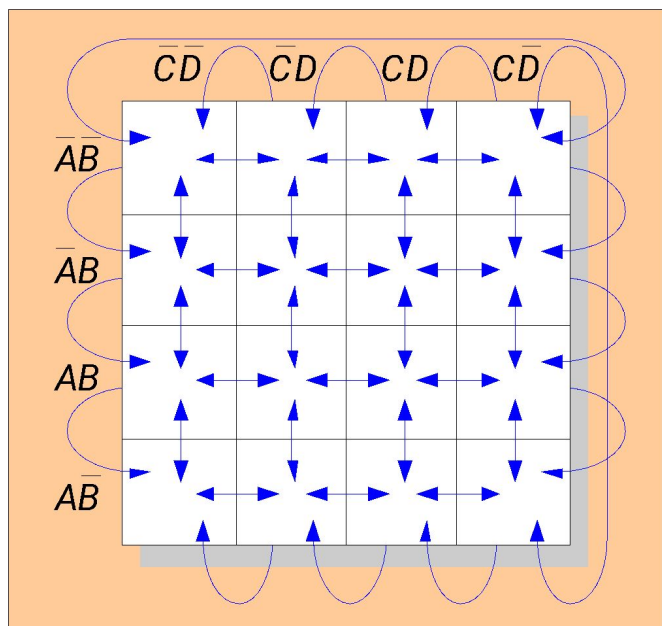
B changes
across this
boundary

C changes
across this
boundary

Summary

Karnaugh maps

A 4-variable map has an adjacent cell on each of its four boundaries as shown.



Each cell is different only by one variable from an adjacent cell.

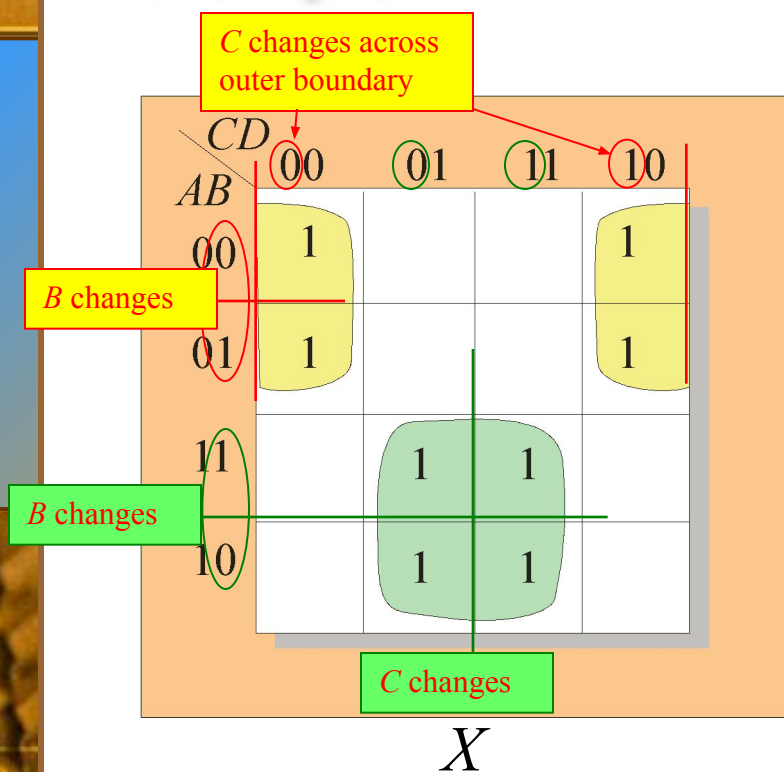
Grouping follows the rules given in the text.

The following slide shows an example of reading a four variable map using binary numbers for the variables...

Summary

Karnaugh maps

Example Group the 1's on the map and read the minimum logic.



Solution

1. Group the 1's into two separate groups as indicated.
2. Read each group by eliminating any variable that changes across a boundary.
3. The upper (yellow) group is read as $\bar{A}\bar{D}$.
4. The lower (green) group is read as AD .

$$X = \bar{A}\bar{D} + AD$$

Quiz

1. The associative law for addition is normally written as

a. $A + B = B + A$

b. $(A + B) + C = A + (B + C)$

c. $AB = BA$

d. $A + AB = A$

Quiz

2. The Boolean equation $AB + AC = A(B + C)$ illustrates

- a. the distribution law
- b. the commutative law
- c. the associative law
- d. DeMorgan's theorem

Quiz

3. The Boolean expression $A \cdot 1$ is equal to

- a. A
- b. B
- c. 0
- d. 1

Quiz

4. The Boolean expression $A + 1$ is equal to

- a. A
- b. B
- c. 0
- d. 1

Quiz

5. The Boolean equation $AB + AC = A(B + C)$ illustrates
- a. the distribution law
 - b. the commutative law
 - c. the associative law
 - d. DeMorgan's theorem

Quiz

6. A Boolean expression that is in standard SOP form is
- a. the minimum logic expression
 - b. contains only one product term
 - c. has every variable in the domain in every term
 - d. none of the above

Quiz

7. Adjacent cells on a Karnaugh map differ from each other by
- a. one variable
 - b. two variables
 - c. three variables
 - d. answer depends on the size of the map

Quiz

8. The minimum expression that can be read from the Karnaugh map shown is

- a. $X = A$
- b. $X = A$ —
- c. $X = B$
- d. $X = B$ —

	$\bar{C}$	C
$\bar{A}\bar{B}$		
$\bar{A}B$		
AB	1	1
$A\bar{B}$	1	1

Quiz

9. The minimum expression that can be read from the Karnaugh map shown is

- a. $X = A$
- b. $X = A$ —
- c. $X = B$
- d. $X = B$ —

	$\bar{C}$	C
$\bar{A}\bar{B}$	1	1
$\bar{A}B$		
AB		
$A\bar{B}$	1	1

Quiz

Answers:

- | | |
|------|------|
| 1. b | 6. c |
| 2. c | 7. a |
| 3. a | 8. a |
| 4. d | 9. d |
| 5. a | |